

Python + Django REST Backend Training Plan

PHASE 1 — Computer Science + Environment

Day 1 — Introduction to Programming & Backend

Topics

- What is programming
- Software development overview
- Frontend vs Backend
- Client-Server architecture
- HTTP request-response cycle
- APIs
- Why Python for backend

Practical

- Install Python
- Install VS Code
- Run first program

```
print("Hello World")
```

Assignment

Write a program to print:

- Your name
 - Your college
 - Your goal (backend developer)
-

Day 2 — Python Environment Setup

Topics

- Python interpreter

- Python versions
- Virtual environment
- pip package manager

Practical

```
python -m venv env
```

```
env\Scripts\activate
```

```
pip install requests
```

Mini Exercise

Call a public API using `requests` .

Assignment

Fetch GitHub user details:

<https://api.github.com/users/<username>>

Print:

- name
 - followers
 - public repos
-

PHASE 2 — Python Fundamentals

Day 3 — Python Syntax & Variables

Topics

- Python syntax
- indentation
- variables
- datatypes
- input / output

Practical

Code block

```
1 name = input("Enter name:")
2 print("Hello", name)
```

Assignment

Create program that takes:

- name
- age
- city

Print formatted sentence.

Day 4 — Operators & Expressions

Topics

- arithmetic operators
- comparison operators
- logical operators

Mini Project

Simple Calculator

Assignment

Create calculator supporting:

+

-

*

/

Day 5 — Conditional Statements

Topics

- if
- if else
- elif

Mini Project

ATM withdrawal system.

Assignment

Check:

- balance
 - withdraw amount
 - insufficient balance
-

Day 6 — Loops

Topics

- for loop
- while loop
- break
- continue

Mini Project

Number guessing game.

Assignment

Generate random number and let user guess.

Day 7 — Data Structures

Topics

- list
- tuple
- set
- dictionary

Mini Project

Student Marks Manager.

Example

Code block

```
1  students = {  
2    "rahu": 90,  
3    "anita": 85  
4  }
```

Assignment

Add / update / delete student marks.

PHASE 3 — Functions

Day 8 — Functions

Topics

- defining functions
- parameters
- return values

Mini Project

Reusable calculator functions.

Assignment

`add()`

`subtract()`

`multiply()`

`divide()`

Day 9 — Advanced Functions

Topics

- `*args`
- `**kwargs`
- `lambda`
- `map`
- `filter`

Assignment

Student marks analyzer:

- `highest`
- `lowest`
- `average`

PHASE 4 — Files & Exceptions

Day 10 — File Handling

Topics

- open
- read
- write
- append
- context manager

Mini Project

Notes manager.

Assignment

Save notes into a file.

Day 11 — Exception Handling

Topics

- try
- except
- finally
- custom exception

Mini Project

Safe calculator.

Assignment

Handle:

- division by zero
 - invalid input
-

PHASE 5 — OOP

Day 12 — Classes & Objects

Topics

- class
- object
- constructor
- instance variables

Mini Project

Student class.

Assignment

Create class:

Student

with:

- name
 - age
 - marks
-

Day 13 — OOP Advanced

Topics

- inheritance
- polymorphism
- encapsulation
- classmethod
- staticmethod

Mini Project

Employee Management System.

Assignment

Create:

Employee

Manager

Developer

using inheritance.

PHASE 6 — Git & GitHub

Day 14 — Version Control

Topics

- version control
- git basics
- GitHub
- repository
- commit

Practical

```
git init
```

```
git add .
```

```
git commit
```

```
git push
```

Assignment

Push all Python projects to GitHub.

PHASE 7 — Database Concepts

Day 15 — Database Fundamentals

Topics

- DBMS
- relational database
- tables
- rows
- primary key
- foreign key

Practical

Install:

- PostgreSQL
- pgAdmin

Assignment

Create database:

student_db

Day 16 — SQL Basics

Topics

- CREATE
- INSERT
- SELECT
- UPDATE
- DELETE
- WHERE
- ORDER BY

Mini Project

Student table.

Assignment

Create table students

Insert 5 records

Update marks

Delete one record

PHASE 8 — Django

Day 17 — Django Introduction

Topics

- Django overview
- MVT architecture

- project structure

Practical

django-admin startproject backend

Run server.

Assignment

Create homepage returning:

Hello Backend Developer

Day 18 — Django App & Models

Topics

- apps
- models
- migrations
- admin panel

Mini Project

Student model.

Assignment

Create model:

Student

name

age

email

Day 19 — PostgreSQL with Django

Topics

- connect postgres
- settings configuration

Assignment

Store students in PostgreSQL database.

Verify using **pgAdmin**.

PHASE 9 — Django REST Framework

Day 20 — REST API Basics

Topics

- REST
- JSON
- HTTP methods
- status codes

Install

`pip install djangorestframework`

Assignment

Create simple API:

GET /hello

Test using **Postman**.

Day 21 — Serializers

Topics

- serializer
- ModelSerializer
- validation

Assignment

Create **Student serializer**.

Day 22 — APIView Basics

Topics

- APIView
- GET method
- POST method

Assignment

Create endpoints:

GET /students

POST /students

Test using **Postman**.

Day 23 — CRUD APIs

Topics

- PUT
- DELETE

Assignment

GET students

POST student

PUT student

DELETE student

All using **APIView**.

Day 24 — Filtering & Search

Topics

- query parameters
- filtering

Example

GET /students?name=rahul

Assignment

Search student by name.

PHASE 10 — Authentication

Day 25 — JWT Authentication

Topics

- authentication vs authorization
- JWT tokens

Install

`pip install djangorestframework-simplejwt`

Assignment

Create APIs:

register

login

token refresh

Day 26 — Protected APIs

Topics

- permissions
- authenticated endpoints

Assignment

Protect:

`POST /students`

Only logged-in users can create.

PHASE 11 — File Upload

Day 27 — File Upload API

Topics

- FileField
- media files

Assignment

Upload student profile picture.

PHASE 12 — Final Backend Project

Day 28–32

Project:

Course Management System

Entities

Users

Courses

Enrollments

Features

Authentication

register

login

JWT

Courses

create course

list courses

update course

delete course

Enrollment

enroll student

view enrolled courses

Test using:

- Postman
- PostgreSQL
- pgAdmin

Students push project to **GitHub**.

Final Skills Students Gain

Python programming

Object-oriented programming

Git & GitHub workflow

SQL & PostgreSQL

Django backend development

Django REST Framework APIs

JWT authentication

API testing with Postman